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FINAL REPORT OF

Whole Transcriptome Sequencing & Analysis

On Illumina Novaseq X Plus Platform

Submitted to:

Dr Madhurjit Singh Rathore

Client Organization

Submitted by:

Unigenome

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NGS-240540 | Date: 09-03-2026

4. Quantity and quality check (QC) of library on Agilent Tape Station 4150
5. Bioinformatics Analysis
6. Data statistics
7. Reference genome information
8. Mapping to reference genome
9. Transcript assembly
0. Differential expression analysis
1. Functional annotation
2. Deliverables

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WARNING

No .fastq.gz files found in 01_Raw_Data (sample detection may rely on Mapping/Stats).

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throughput technology to provide deep coverage of the transcriptome.

Project Metadata

Q

Search rows...

FIELD	VALUE
Project PI	Dr Madhurjit Singh Rathore
Application	WTA
No. of Samples	4

Table 1.1: Project Specifications

SERVICE TYPE	PLATFORM	READ LENGTH	DATA THROUGHPUT
Transcriptome Sequencing	Illumina Novaseq X Plus	2 X 150 PE	~05 GB / Sample

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under strict temperature controls during transport.

Table 2.1: Sample Overview

TYPE	NO. OF SAMPLES	SHIPPING CONDITION	LIBRARIES PREPARED
Plant Root	4	-20°C	4

Sample Nomenclature: NGS_240540_PM-CONTROL, NGS_240540_PM-CONTROL-2, NGS_240540_PM-DROUGHT, NGS_240540_PM-DROUGHT-1

Total RNA was extracted using the QIAGEN RNeasy mini kit. The quantity was measured using Qubit® 4.0 Fluorometer, and RNA integrity (RIN) was assessed via Agilent 4150 TapeStation to ensure high-quality input material (RIN > 7.0).

 3.2 Preparation of Library

Sequencing libraries were prepared using the NEBNext® Ultra™ RNA Library Prep Kit. Ribosomal RNA was depleted using the KAPA RNA HyperPrep Kit with RiboErase (HMR) to maximize the coverage of coding and non-coding transcripts.

3.3 Quantification using Qubit 4.0 Fluorometer

Table 3.1: RNA Quantification (Qubit)

S.N	SAMPLE ID	CONC. (NG/ML)	VOLUME (ML)	YIELD (MG)	REMARKS
No Qubit data available					

3.4 Cluster Generation and Sequencing

After obtaining the Qubit concentration for the library and the mean peak size from Tape Station profile, library will be loaded onto illumine Novaseq X Plus for cluster generation and sequencing. Paired-End sequencing allows the template fragments to be sequenced in both the forward and reverse directions. The library molecules will bind to complementary adapter oligos on paired-end flow cell. The adapters are designed to allow selective cleavage of the forward strands after re-synthesis of the reverse strand during sequencing. The copied reverse strand is then used to sequence from the opposite end of the fragment.

sizes and ensure proper fragment distribution.

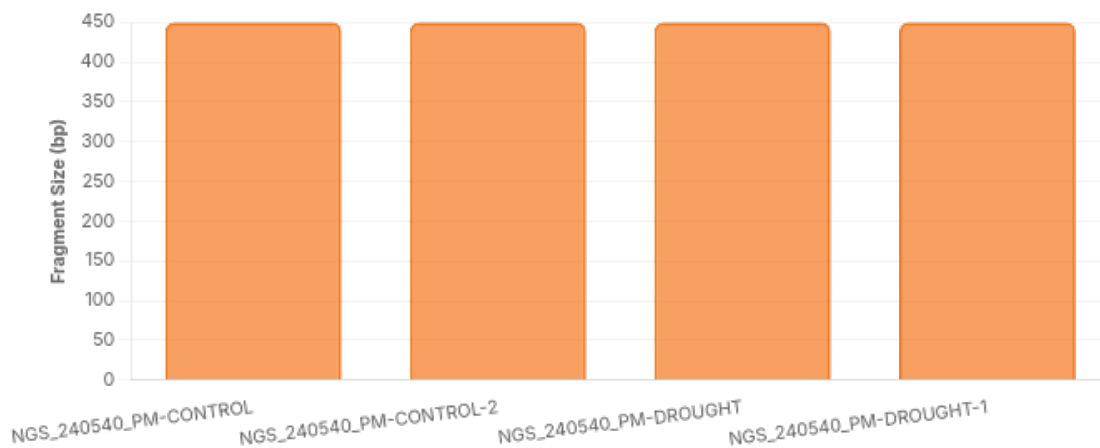


Figure 4.1: Mean library peak size distribution (bp) validated via TapeStation 4150.

4.1 Wet Lab Inferences

The average sizes of the generated libraries were determined to be 450, 450, 450, 450 respectively for the 4 samples. All libraries met the concentration and size requirements for high-quality sequencing on the Novaseq X Plus platform.

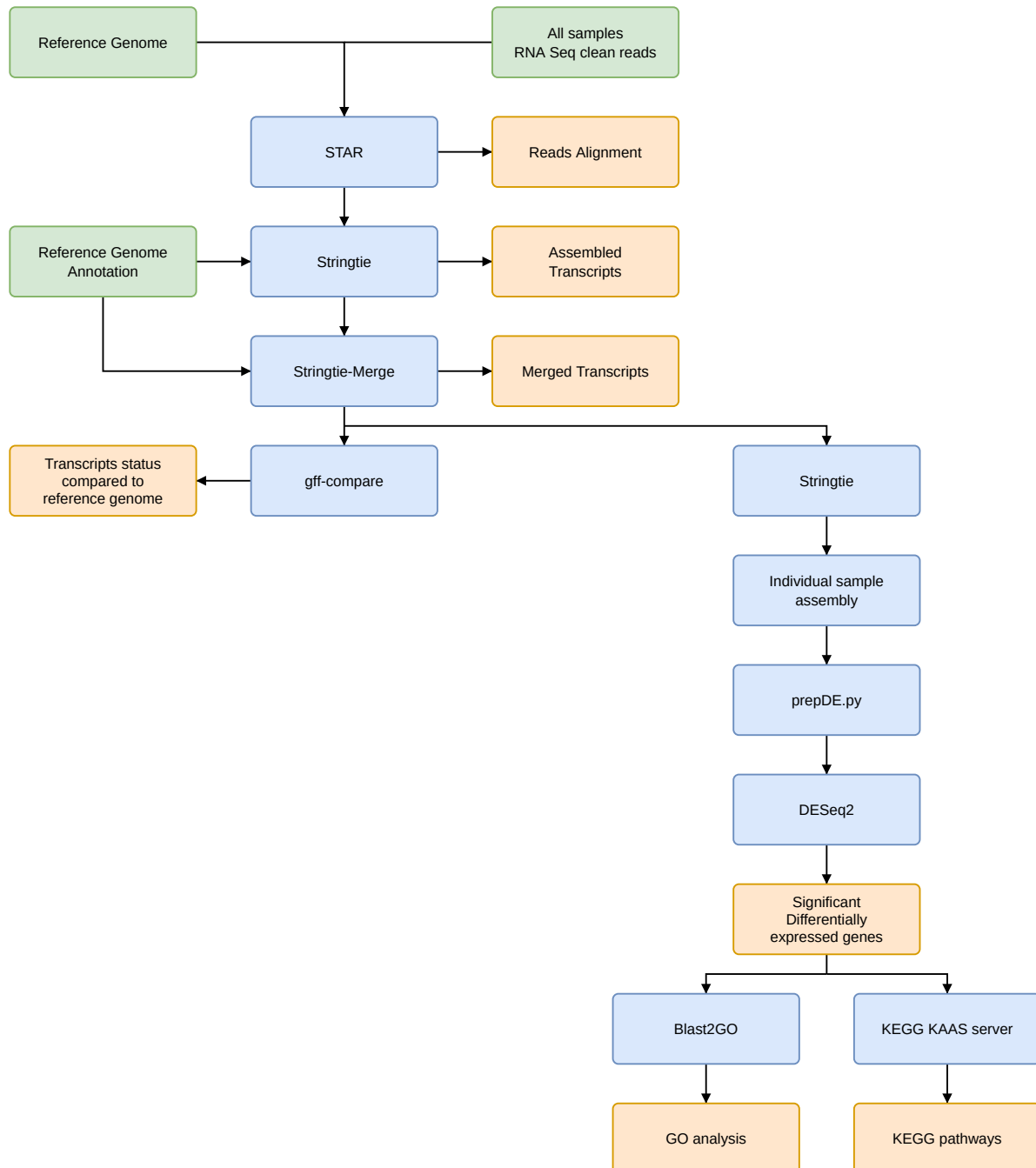


Figure 6: Bioinformatics Analysis Workflow

Reference guided transcript assembly was performed for samples first by mapping clean reads on reference genome using STAR (v2.7.10a) aligner and then performing transcript assembly by StringTie (v 2.2.1). A consensus set of transcripts was obtained using StringTie merge function which merges together all the gene structures found in any of the samples. Transcript abundance was then estimated for individual sample using merged transcript consensus again using StringTie and read counts thus obtained for each transcript were taken as input for differential expression analysis using DESeq2 package. Gene Ontology and pathway analysis of the significantly differentially expressed transcripts were performed using Blast2go cli and KEGG-KAAS server respectively. Overall bioinformatics workflow is graphically represented in figure above and software version along with its use in table below.

5.2 Detail of Software used for analysis

Table 5.1: Software Stack

SOFTWARE	VERSION	APPLICATION
Trim Galore	0.6.4	Adapter and low-quality data removal
STAR	2.7.10	Mapping of reads to reference genome
Stringtie	2.2.1	Reference based assembly
gffcompare	0.12.6	Comparing reference genome gff with merged gff
gffread	0.12.7	Fetching fasta sequences from gff
Blastx	2.30.0+	Similarity search against NCBI NR
Blast2go cli	1.5	GO mapping and annotation
edgeR	3.6.2	Differential expression analysis
KEGG KAAS	Webserver	Pathway analysis
clusterProfiler	4.12.6	Over-representation analysis

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Below:

TOTAL SAMPLES
4

AVG. DATA/SAMPLE
~5.2 GB

PLATFORM
Novaseq X Plus

Table 6.1: Sequencing Data Statistics

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Search rows...

SAMPLE	TOTAL READS IN R1	TOTAL READS IN R2	TOTAL READS (R1+R2)	TOTAL BASES	DATA (GB)
PM-CONTROL-2	N/A	N/A	52,580,666	8,360,325,894	8.36
PM-CONTROL	N/A	N/A	59,173,242	9,408,545,478	9.41
PM-DROUGHT-1	N/A	N/A	48,744,634	7,750,396,806	7.75
PM-DROUGHT	N/A	N/A	49,981,628	7,947,078,852	7.95

Genomes/any/001/000/012/010/001-000012201-00001217/

The reference GFF file contains genome annotation. It includes information (locus and description) regarding genes and proteins. There are a total of 40021 genes and 44391 mRNAs as per the GFF file of NCBI reference genome. The fasta sequence of the reference genome along with the corresponding GFF file is provided with the deliverables in the folder named '02_reference_genome_and_gff' for client's reference.

Table 7.1: Reference Genome Statistics

Q

Search rows...

DESCRIPTION	STATS
Total Chromosomes/Scaffolds	7
Total Genome Length (bp)	1,564,537,551
Mean Scaffold Size	223,505,364
Max Scaffold Size (bp)	300,907,625
Total Genes Annotated	0

the indexed reference genome generated in previous step was given to the STAR aligner. The analysis begins by mapping NG reads against the reference genome to identify the positions from where the reads originated. This mapping information allows us to collect subsets of the reads corresponding to each gene, and then to assemble and quantify transcripts represented by those reads. Hence, the high-quality reads were mapped to reference genome using STAR aligner to create alignment in BAM format for each sample with default parameters. The mapping statistics are provided in the table below:

Table 8.1: Mapping Statistics

Q

Search rows...

SAMPLE NAME	TOTAL CLEAN READS	NO. OF MAPPED READS	% MAPPED	# UNIQUELY MAPPED	% UNIQUELY MAPPED
NGS_240540_PM- CONTROL	N/A	25,719,449	87.1	24,904,947	84.35
NGS_240540_PM- CONTROL-2	N/A	23,075,339	87.97	22,356,107	85.23
NGS_240540_PM- DROUGHT	N/A	21,864,503	87.69	21,167,251	84.89
NGS_240540_PM- DROUGHT-1	N/A	21,246,476	87.37	20,575,828	84.61

assembling each reads into as many isoforms as are needed to explain the data. Following this, StringTie simultaneously assembles and quantify the final transcripts by using network flow algorithm and starting from most highly abundant transcripts. The reference genome GFF annotation files, containing exon structures of 'known' genes, are then used to annotate the assembled transcripts and quantify the expression of known genes as well derive clues if a novel transcript has been found in the sample.

After assembling each sample, the full set of assemblies is passed to StringTie's merge function, which merges together all the gene structures found in any of the samples. This step is required because transcripts in some of the samples might only be partially covered by reads, and as a consequence only partial versions of them will be assembled in the initial StringTie run. The merge step creates a set of transcripts that is consistent across all samples, so that the transcripts can be compared in subsequent steps in figure below:

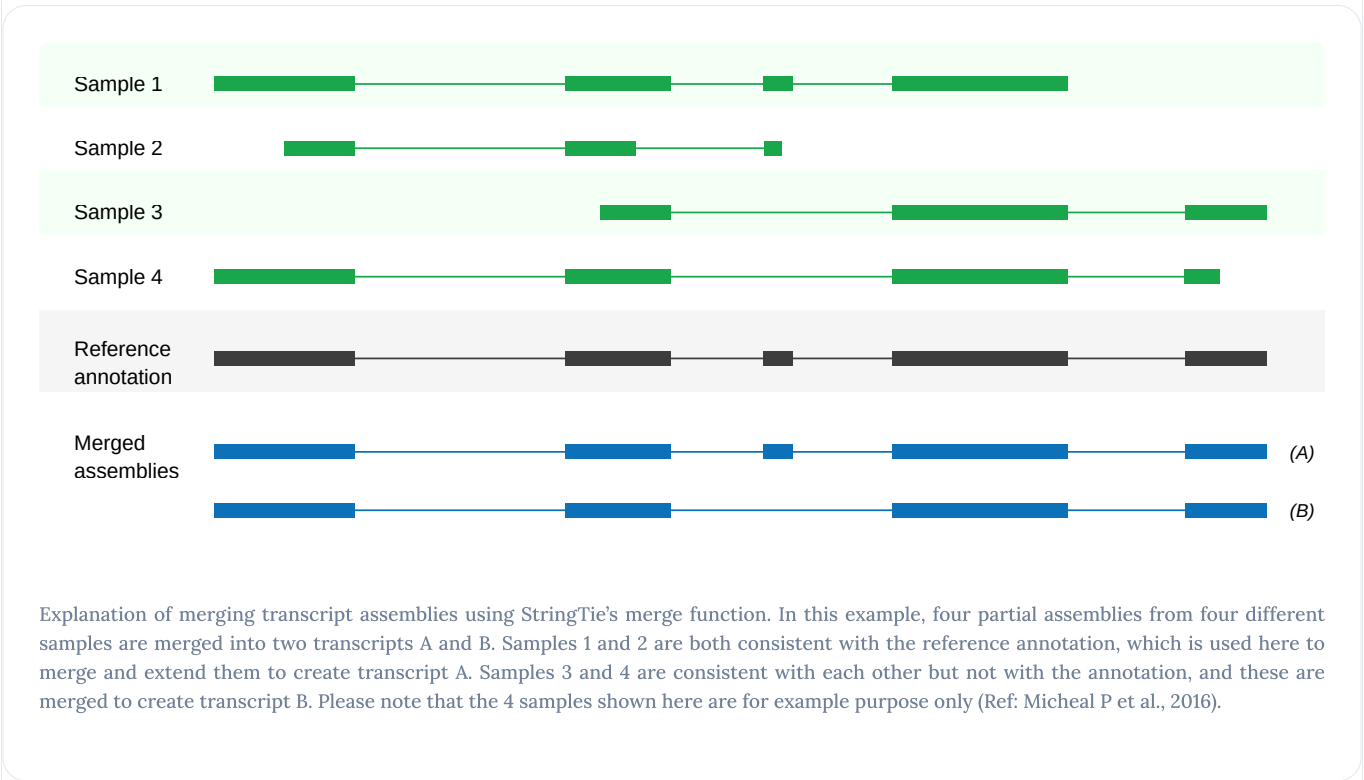


Table 9.1: Assembly Statistics

Q Search rows...				
SAMPLE NAME	# ASSEMBLED TRANSCRIPTS	TOTAL (BP)	MEAN SIZE (BP)	
merged.fasta	66,758	92,931,869	1,392.07	
PM-CONTROL	45,703	72,429,977	1,584.80	
PM-DROUGHT-1	45,820	72,559,758	1,583.58	
PM-CONTROL-2	45,710	72,393,952	1,583.77	
PM-DROUGHT	45,288	71,722,434	1,583.70	

=	complete, exact match of intron chain	o	other same strand overlap with reference exons
c	contained in reference (intron compatible)	s	intron match on the opposite strand (likely a mapping error)
k	containment of reference (reverse containment)	x	exonic overlap on the opposite strand (like o or e but on the opposite strand)
m	retained intron(s), all introns matched or retained	i	fully contained within a reference intron
n	retained intron(s), not all introns matched/covered	y	contains a reference within its intron(s)
j	multi-exon with at least one junction match	p	possible polymerase run-on (no actual overlap)
e	single exon transfrag partially covering an intron, possible pre-mRNA fragment	r	repeat (at least 50% bases soft-masked)
		u	none of the above (unknown, intergenic)

Figure 9.1: Different class codes and their descriptions (<https://ccb.jhu.edu/software/stringtie/gffcompare.shtml>)

9.2 Identification of novel isoform transcripts

To identify novel isoform transcripts not present in the reference GTF file, gffcompare utility was ran taking the reference GTF and the string-tie merged GTF file. This produces an output file, which adds to each transcript a 'class code' and the name of the transcript from the reference annotation file to check how the predicted transcripts relate to an annotation file. Class code 'j' means that predicted transcript is a potential novel isoform that shares at least one splice junction with a reference transcript. A total of 8,934 novel isoforms with the class code 'j' were extracted and are provided with the deliverables in a file named novel.isoforms.gtf in the folder '03_transcript_assembly_gtf'. A brief description of the column names in the GTF file is provided in the same folder in the file named ReadMe.txt.

Step: A python program (prepDE.py) was used to extract the read count information directly from the files generated by StringTie. For differential expression analysis, sample comparison was made according to grouping information provided by client:

Comparison information

Table 10.1: Comparison information

Q

Search rows...

COMPARISON	DESCRIPTION
ST v/s SC	Comparison 1 [ST (ST1+ST2) Vs SC (SC1+SC2)]
RC v/s SC	Comparison 2 [RC (RC1+RC2) Vs SC (SC1+SC2)]
RT v/s SC	Comparison 3 [RT (RT1+RT2) Vs SC (SC1+SC2)]
RT v/s ST	Comparison 4 [RT (RT1+RT2) Vs ST (ST1+ST2)]
RC v/s ST	Comparison 5 [RC (RC1+RC2) Vs ST (ST1+ST2)]
RT v/s RC	Comparison 6 [RT (RT1+RT2) Vs RC (RC1+RC2)]

Group wise sample information

Table 10.2: Sample description

Q

Search rows...

GROUP NAME	SAMPLE NAME
SC	SC1
SC	SC2
ST	ST1
ST	ST2
RC	RC1
RC	RC2
RT	RT1
RT	RT2

Differential gene expression was inferred between samples by applying the R package DESeq2. It is bioconductor package based on negative binomial distribution method. The analysis provides tabular result along with the normalized i.e. "baseMean" for each involved sample. The description of selected result columns has been given in below mentioned table:

NAME	
	Feature identifier
baseMean	The average of the normalized count values, dividing by size factors, taken over all samples
log2FoldChange	The effect size estimate. This value indicates how much the gene or transcript's expression seems to have changed between the comparison and control groups. This value is reported on a logarithmic scale to base 2.
lfcSE	The standard error estimate for the log2 fold change estimate.
stat	The value of the test statistic for the gene or transcript.
pvalue	P-value of the test for the gene or transcript.
padj	Adjusted P-value for multiple testing for the gene or transcript.

The criterion used to identify upregulated and downregulated transcripts along with the significance is provided in the below table:

Table 10.4: Criteria used to identify upregulated and downregulated genes and assigning the significance

Search rows...	
CONDITION	STATUS
log2FoldChange > 0	Up regulated
log2FoldChange < 0	Down regulated
log2FoldChange > 0 and padj <0.05	Significantly up regulated
log2FoldChange < 0 and padj <0.05	Significantly down regulated

Differential expression analysis statistics is provided in table below and the data deliverable are given in the folder "05_differential_expression_analysis" which contains the Differential gene expression (DGE) analysis results.

Table 10.5: Differential gene expression statistics

COMPARISON	TOTAL DEGS	SIG. DOWN	SIG. UP	TOTAL SIGNIFICANT
Comparison2	27819	1349	1220	2569
Comparison1	28070	966	803	1769

Annotation for differentially expressed transcripts was obtained based on blastX result by similarity search against NCBI's NR database in order to obtain gene ontology analysis using blast2GO.

10.1 MA Plot for differentially expressed genes

The plot visualizes the differences between measurements taken in two samples, by transforming the data onto M (log ratio) and A (mean average) scales, then plotting these values. For illustration purpose, MA plot for Comparison1 has been depicted below:

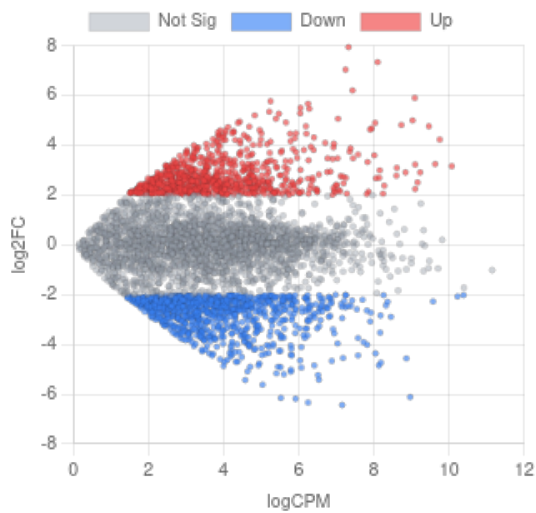
Select a comparison to view interactive MA and Volcano plots.

Comparison1

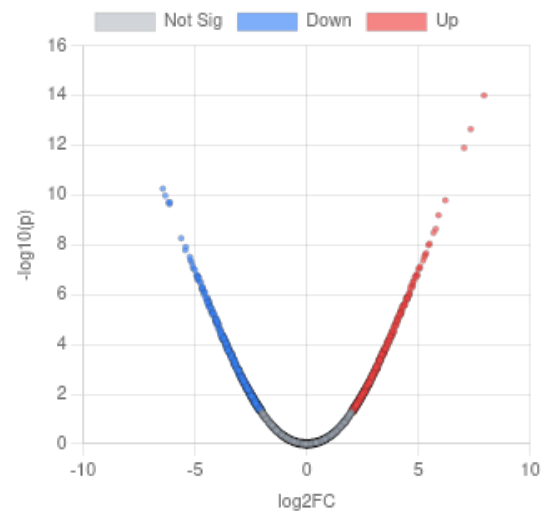
Comparison2

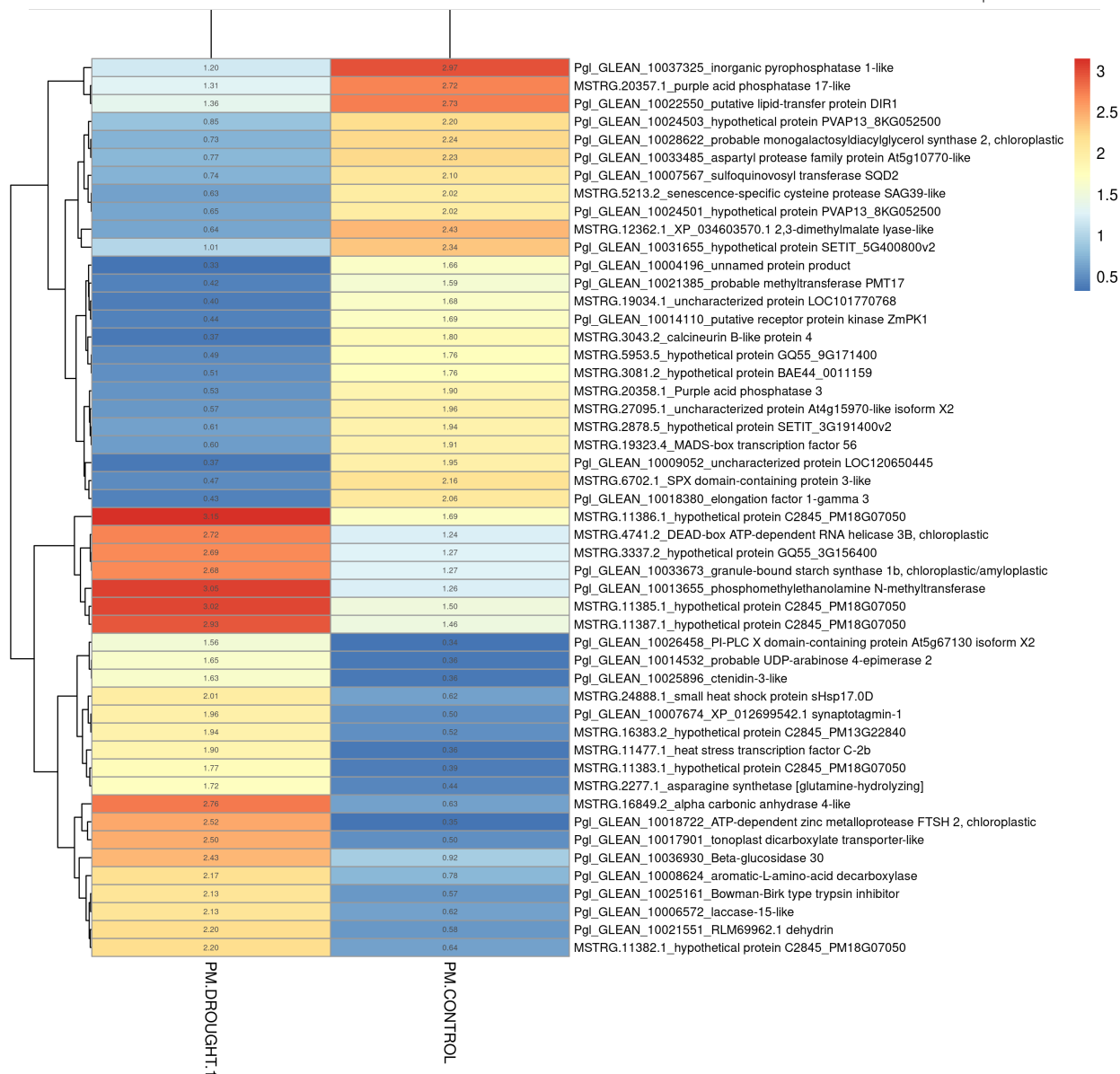
Comparison: Comparison1

MA Plot



Volcano Plot





Heatmap of significantly differentially expressed genes for comparison1

The Gene Ontology project provides controlled vocabularies of defined terms representing gene product properties. These cover three domains: Cellular Component, the parts of a cell or its extracellular environment; Molecular Function, the elemental activities of a gene product at the molecular level, such as binding or catalysis; and Biological Process, operations or sets of molecular events with a defined beginning and end, pertinent to the functioning of integrated living units: cells, tissues, organs, and organisms. GO was assigned to significant differentially expressed transcripts using Blast2go cli. Single gene can be assigned with multiple GO categories and hence multiple GO terms. The GO domain distribution is shown in table below:

Table 11.1: GO Distribution of Significantly Differential expression transcripts

Q

Search rows...

SAMPLE NAME	SIGNIFICANT DGE	# SEQ WITH GO	BIOLOGICAL PROCESS	CELLULAR COMPONENT	MOLECULAR FUNCTION
Comparison1	1357	498	398	351	412
Comparison2	4616	1492	1178	1022	1189
Comparison3	5601	1910	1475	1311	1534
Comparison4	4916	1551	1194	1036	1214
Comparison5	5331	1805	1410	1229	1433
Comparison6	2793	1215	982	879	981

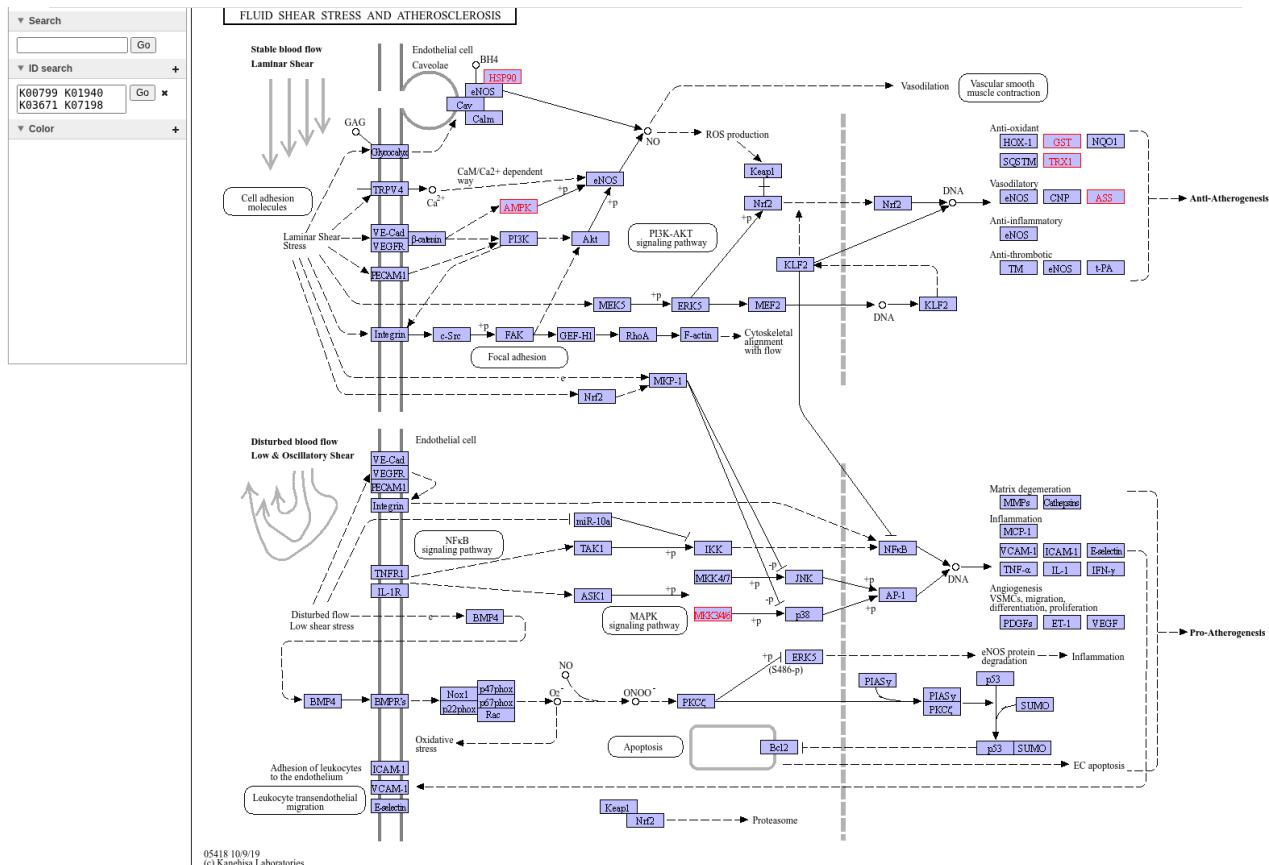
Assigned Gene Ontology are also provided in deliverables "06_Significant_DGE_GO".

11.2 Pathway analysis for significant DGE

Mapping of significantly differentially expressed transcripts to biological pathways was performed using **KEGG-KAAS**. Results represent metabolic pathways of major biomolecules.

Table 11.2: KEGG Pathway Statistics

LEVEL 1 CATEGORY	LEVEL 2 SUB-CATEGORY	COUNTS
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Representative KEGG pathway image

KEGG Enrichment (preview)

Open KEGG enrichment results (XLSX)

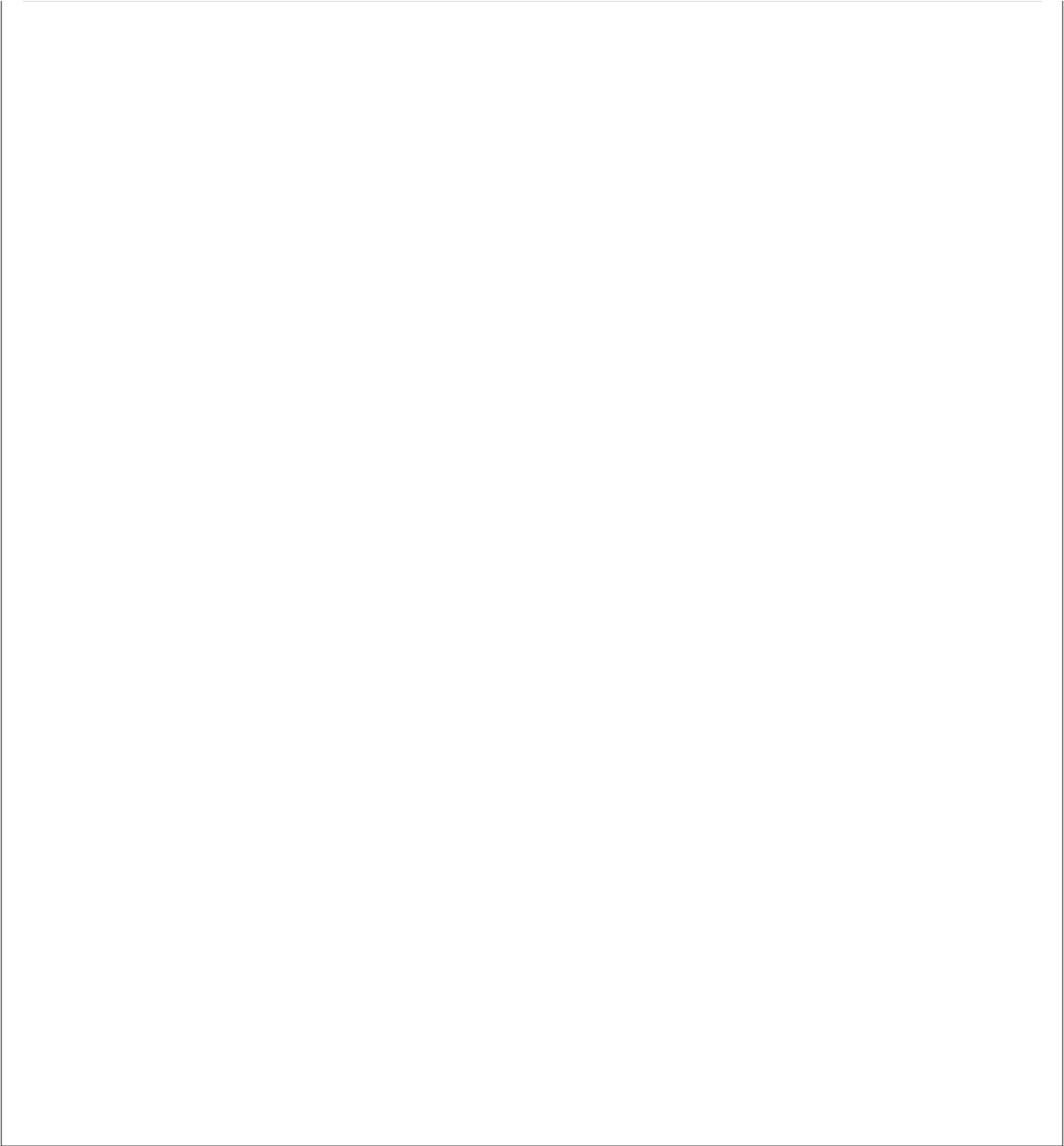
Top enriched KEGG pathways (first 10 rows)

ID	DESCRIPTION	GENERATIO	BGRATIO	PVALUE	P.ADJUST	QVALUE	COUNT
sbi00500	Starch and sucrose metabolism	18/232	169/5611	0.000187701398466087	0.0206471538312696	0.0203507832021126	18

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GeneRatio

Representative enrichment plot

- Raw Data (FastQ)
- Clean Data
- Reference Genome & GFF
- BAM Alignment Files
- Gene Count Matrix
- Differential Expression Results
- GO & KEGG Enrichment Results

Project Directory Structure

```
├── 01_Raw_Data
│   ├── NGS_240540_md5sum.txt
│   ├── NGS_240540_Raw_Stats.txt
│   ├── PM-CONTROL-2_S73_L008_R1_001.fastq.gz
│   ├── PM-CONTROL-2_S73_L008_R2_001.fastq.gz
│   ├── PM-CONTROL_S74_L008_R1_001.fastq.gz
│   ├── PM-CONTROL_S74_L008_R2_001.fastq.gz
│   ├── PM-DROUGHT-1_S72_L008_R1_001.fastq.gz
│   ├── PM-DROUGHT-1_S72_L008_R2_001.fastq.gz
│   ├── PM-DROUGHT_S71_L008_R1_001.fastq.gz
│   └── PM-DROUGHT_S71_L008_R2_001.fastq.gz
├── 02_reference_genome_and_gff
│   ├── pearl_millet_v1.1.fa
│   └── PM.genechr.trans.gff
├── 03_transcript_assembly_gtf
│   ├── gffcompare_results.gtf
│   ├── merged_transcripts.gtf
│   ├── NGS_240540_PM-CONTROL-2Aligned_1.gtf
│   ├── NGS_240540_PM-CONTROLAligned_1.gtf
│   ├── NGS_240540_PM-DROUGHT-1Aligned_1.gtf
│   ├── NGS_240540_PM-DROUGHTAligned_1.gtf
│   ├── novel.isoforms.gtf
│   └── ReadMe.txt
├── 04_transcript_sequences_fasta
│   ├── Merged.fasta
│   ├── NGS_240540_PM-CONTROL-2Aligned_transcript.fasta
│   ├── NGS_240540_PM-CONTROLAligned_transcript.fasta
│   ├── NGS_240540_PM-DROUGHT-1Aligned_transcript.fasta
│   └── NGS_240540_PM-DROUGHTAligned_transcript.fasta
├── 05_differential_expression_analysis
│   ├── Comparison1_DGE.xlsx
│   ├── Comparison1_Heatmap.png
│   ├── Comparison1_MA_and_Volcano.pdf
│   ├── Comparison2_DGE.xlsx
│   ├── Comparison2_Heatmap.png
│   └── Comparison2_MA_and_Volcano.pdf
├── 06_Significant_DGE_GO
│   ├── Comparison1_Significant_DGE_with_GO.xlsx
│   └── Comparison2_Significant_DGE_with_GO.xlsx
├── 07_Significant_DGE_pathways
│   ├── Comparison1_Significant_DGE_Pathway_images
│   ├── Comparison1_Significant_DGE_Pathways.xlsx
│   ├── Comparison2_Significant_DGE_Pathway_images
│   └── Comparison2_Significant_DGE_Pathways.xlsx
├── 08_Significant_DGE_Enrichment
│   ├── C1_Sig_down_barplot_GO_enrichment.png
│   ├── C1_Sig_down_barplot_KEGG_enrichment.png
│   ├── C1_Sig_down_dotplot_GO_enrichment.png
│   └── C1_Sig_down_dotplot_KEGG_enrichment.png
```

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```
| └─ C1_Sig_up_G0_enrichment_results.xlsx
| └─ C1_Sig_up_KEGG_enrichment_results.xlsx
| └─ C2_Sig_down_barplot_G0_enrichment.png
| └─ C2_Sig_down_barplot_KEGG_enrichment.png
| └─ C2_Sig_down_dotplot_G0_enrichment.png
| └─ C2_Sig_down_dotplot_KEGG_enrichment.png
| └─ C2_Sig_down_G0_enrichment_results.xlsx
| └─ C2_Sig_down_KEGG_enrichment_results.xlsx
| └─ C2_Sig_up_barplot_G0_enrichment.png
| └─ C2_Sig_up_barplot_KEGG_enrichment.png
| └─ C2_Sig_up_dotplot_G0_enrichment.png
| └─ C2_Sig_up_dotplot_KEGG_enrichment.png
| └─ C2_Sig_up_G0_enrichment_results.xlsx
| └─ C2_Sig_up_KEGG_enrichment_results.xlsx
└─ Readme.txt
```